

# Volume–Volatility Spike Signal in Equities

A Cross-Sector Backtesting Study: 2015–2026

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## Abstract

This paper surges as an independent study of the relation between trading volume, rolling volatility, and risk-adjusted across eight publicly traded companies from 2015 to 2026. Using Python and daily OHLCV data sourced from Yahoo Finance, we compute 20 day rolling metric and calculate the correlation between volume and volatility across different market types and trends. We develop a signal that triggers when both rolling volatility and volume simultaneously exceed their 20-day moving averages. Backtesting this signal more than 3000 instances result in consistent positive forward returns at 5, 10, and 20 day horizons in 7 out of 8 companies tested, with 10 day returns ranging from 0.5% to 3.4% and win rates of 52-62%. The signal only failed for Boeing (BA), a over-all declining company, a result that suggests the strategy captures institutional accumulation behaviour specific to fundamentally sound businesses.

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## 1 Introduction

The trading volume and price volatility relation is one the most studied subjects in the market microstructure. The conventional view holds that volume and volatility are positively correlated, large price moves attract more participants, and more participants generate larger price moves. However, the directional implication of this relationship is not-so-well understood as when both volume and volatility spike simultaneously, does this signal a continuation or a reversal?

The signal proposed in this paper is a simple, interpretable, based on the joint spike and tests its predictive power across eight equities from four different sectors over an eleven-year period. The signal requires no external data, no machine learning, and no proprietary inputs, only publicly available daily OHLCV data.

The motivation is grounded in behavioural finance. Kahneman and Tversky's (1979) prospect theory documents that investors respond more strongly to losses than to equivalent gains — loss aversion. We hypothesise that volume-volatility spikes disproportionately reflect fear-driven selling, and that the subsequent price recovery generates the observed positive forward returns.

## 2 Methodology

### 2.1 Data

The daily OHLCV data was downloaded using the `yfinance` Python library for eight tickers across four sectors:

- **Semiconductors:** MU (Micron), NVDA (NVIDIA), AMD, INTC (Intel)
- **Financials:** JPM (JPMorgan), GS (Goldman Sachs)
- **Industrials:** BA (Boeing)
- **Mining:** SQM

The sample period runs from **January 2015 to June 2026**, providing approximately 2,800 trading days per ticker.

### 2.2 Rolling Metrics

Daily log returns are computed as:

$$r_t = \frac{P_t - P_{t-1}}{P_{t-1}} \quad (1)$$

The 20-day rolling volatility  $\sigma_{20,t}$  and rolling mean  $\mu_{20,t}$  are:

$$\sigma_{20,t} = \sqrt{\frac{1}{19} \sum_{i=0}^{19} (r_{t-i} - \mu_{20,t})^2} \quad (2)$$

$$\mu_{20,t} = \frac{1}{20} \sum_{i=0}^{19} r_{t-i} \quad (3)$$

The annualised rolling Sharpe ratio is computed as:

$$\text{Sharpe}_{20,t} = \frac{\mu_{20,t} - r_f/252}{\sigma_{20,t}} \cdot \sqrt{252} \quad (4)$$

where  $r_f = 3.5\%$  is the assumed annual risk-free rate.

### 2.3 Signal Definition

The signal fires on day  $t$  when both rolling volatility and volume exceed their respective 20-day moving averages simultaneously:

$$\mathcal{S}_t = \mathbf{1} [\sigma_{20,t} > \bar{\sigma}_{20,t}] \cap \mathbf{1} [V_t > \bar{V}_{20,t}] \quad (5)$$

where  $\bar{\sigma}_{20,t}$  and  $\bar{V}_{20,t}$  denote the 20-day moving averages of rolling volatility and volume respectively.

## 2.4 Forward Return Calculation

For each signal date  $t$ , the  $n$ -day forward return is:

$$R_{t,n} = \frac{P_{t+n} - P_t}{P_t} \quad (6)$$

We compute  $R_{t,n}$  for  $n \in \{5, 10, 20\}$  trading days. The win rate is defined as the proportion of signal instances with positive forward return:

$$\text{Win Rate}_n = \frac{1}{|S|} \sum_{t \in S} \mathbf{1}[R_{t,n} > 0] \quad (7)$$

## 2.5 Volume–Volatility Correlation

The Pearson correlation between daily volume  $V_t$  and rolling volatility  $\sigma_{20,t}$  is computed over the full sample, and separately for days when volatility was rising ( $\Delta\sigma_{20,t} > 0$ ) versus falling ( $\Delta\sigma_{20,t} < 0$ ).

# 3 Results

## 3.1 Volume–Volatility Correlation

Table 1 reports the Pearson correlation between volume and rolling volatility across all tickers.

Table 1: Volume–Volatility Pearson Correlation (2015–2026)

| Ticker | Sector        | Overall | Vol Rising | Vol Falling |
|--------|---------------|---------|------------|-------------|
| MU     | Semiconductor | 0.317   | 0.330      | 0.298       |
| NVDA   | Semiconductor | 0.314   | 0.312      | 0.330       |
| AMD    | Semiconductor | 0.154   | 0.169      | 0.114       |
| INTC   | Semiconductor | 0.608   | 0.607      | 0.632       |
| JPM    | Financial     | 0.415   | 0.424      | 0.426       |
| GS     | Financial     | 0.286   | 0.317      | 0.244       |
| BA     | Industrial    | 0.698   | 0.703      | 0.709       |
| SQM    | Mining        | 0.308   | 0.304      | 0.319       |

Correlations range from 0.154 (AMD) to 0.698 (BA), indicating that while volume and volatility are positively related across all tickers, the strength of this relationship varies considerably by company.

## 3.2 Backtest Results

Table 2 presents the full backtest results.

Table 2: Signal Backtest Results — Forward Returns and Win Rates (2015–2026)

| Ticker      | N   | Avg Forward Return |        |        | Win Rate |       |       |
|-------------|-----|--------------------|--------|--------|----------|-------|-------|
|             |     | 5d                 | 10d    | 20d    | 5d       | 10d   | 20d   |
| MU          | 615 | +0.69%             | +2.20% | +2.83% | 52.4%    | 57.8% | 54.2% |
| NVDA        | 583 | +1.66%             | +3.40% | +4.55% | 60.5%    | 62.5% | 61.8% |
| AMD         | 623 | +0.75%             | +1.82% | +4.42% | 53.6%    | 56.0% | 56.7% |
| INTC        | 603 | +0.61%             | +1.54% | +2.85% | 52.1%    | 52.9% | 55.1% |
| JPM         | 622 | +0.28%             | +0.51% | +1.70% | 59.3%    | 58.7% | 63.4% |
| GS          | 624 | +0.29%             | +0.78% | +1.70% | 55.7%    | 58.6% | 56.7% |
| BA          | 581 | −0.21%             | −0.07% | +0.69% | 47.1%    | 55.4% | 55.2% |
| SQM         | 620 | +0.75%             | +1.04% | +2.04% | 52.8%    | 56.6% | 54.0% |
| Avg (ex-BA) | —   | +0.72%             | +1.61% | +2.87% | 55.2%    | 57.6% | 57.4% |

The 10-day horizon consistently yields the strongest win rates across tickers, peaking at 62.5% for NVDA. Boeing is the sole exception, generating negative 5-day returns and a below-random 47.1% win rate at the 5-day horizon.

## 4 Discussion

### 4.1 Why the Signal Works

The consistent positive forward returns across 7 of 8 tickers suggest the signal captures a systematic market dynamic. We propose two complementary explanations:

**Institutional accumulation.** Volume-volatility spikes often follow negative news or market corrections. Institutional investors — who dominate volume in liquid equities — use these dislocations to accumulate positions at attractive prices. The subsequent price recovery reflects this demand.

**Loss aversion overshoot.** Consistent with Kahneman and Tversky (1979), retail investors disproportionately panic-sell during volatility spikes, temporarily pushing prices below fair value. The subsequent mean reversion generates the observed forward returns.

### 4.2 Why Boeing Fails

Boeing’s negative 5-day signal return reflects a fundamentally different regime. Volume-volatility spikes in BA during 2015–2026 were predominantly associated with structural deterioration — the 737 MAX crisis, COVID-related collapse, labour strikes, and regulatory failures. In this context, the signal correctly identifies heightened activity but that activity reflects genuine fundamental decline rather than a temporary dislocation. The signal appears to work only when the underlying business has sound long-term fundamentals.

### 4.3 Limitations

1. **In-sample only.** All results are in-sample. Out-of-sample validation (e.g., training on 2015–2020 and testing on 2021–2026) is required before drawing strong conclusions.
2. **Bull market bias.** The 2015–2026 period was broadly bullish, particularly for semiconductors. Signal performance in extended bear markets is untested.
3. **Transaction costs.** No transaction costs, slippage, or market impact have been modelled. These would reduce net returns, particularly for higher-frequency horizons.
4. **Single signal.** The signal uses only volume and volatility. Combining with additional factors (RSI, earnings calendar, sector momentum) may improve precision.

## 5 Conclusion

This paper presents a simple, interpretable trading signal based on simultaneous spikes in rolling volatility and volume. Backtested across 3,871 signal instances from 2015 to 2026, the signal generates consistent positive forward returns in 7 of 8 companies tested, with an average 10-day return of 1.61% and win rate of 57.6% excluding Boeing.

The failure of the signal on Boeing — and its success on fundamentally sound businesses across multiple sectors — suggests the strategy captures institutional accumulation behaviour following temporary dislocations in quality companies, consistent with behavioural finance theory.

**Next steps** include out-of-sample validation, transaction cost modelling, position sizing optimisation using the Markowitz framework, and extension to additional tickers and asset classes.

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**Code:** [github.com/Santiibabio/Quant-Strategies](https://github.com/Santiibabio/Quant-Strategies)

*This research was conducted independently for educational purposes. Nothing herein constitutes investment advice.*